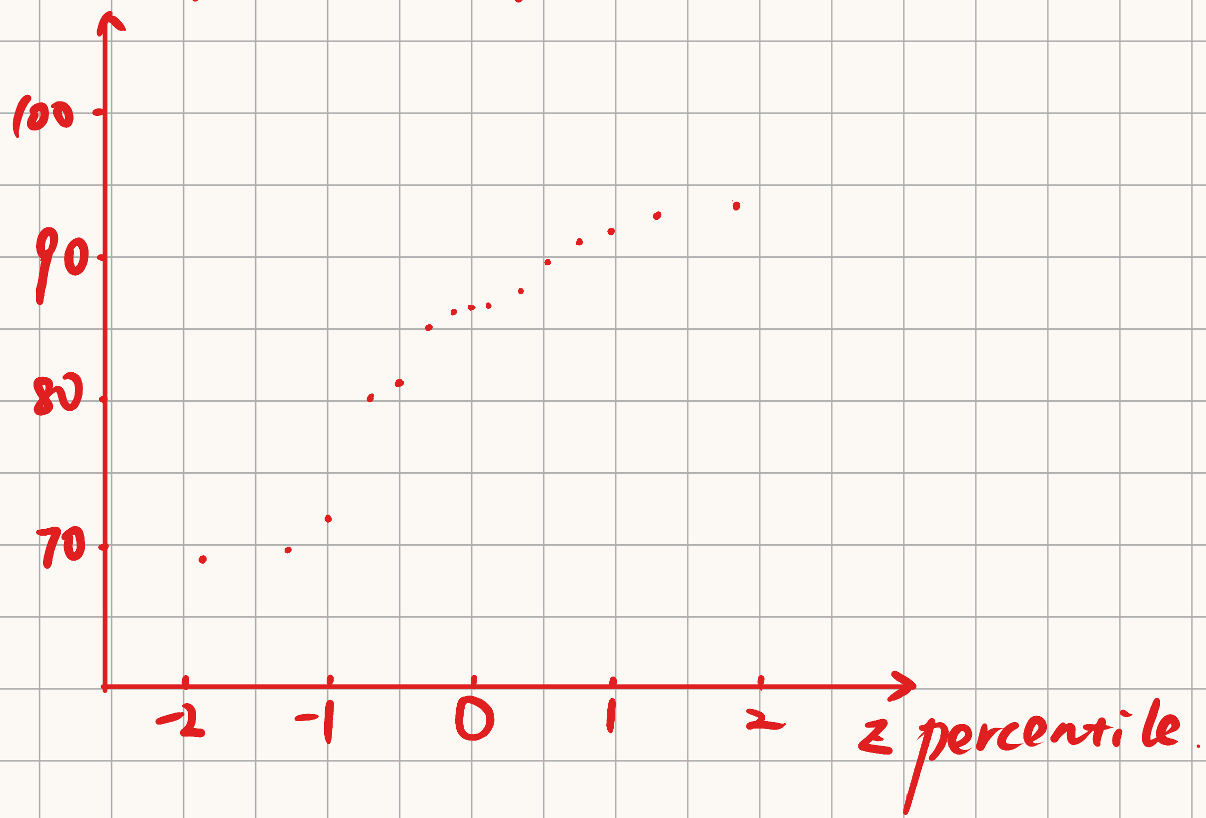
Section 4.6

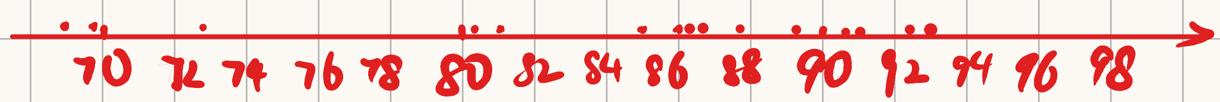
87. Since the plots appears the trend of a line, it shows the tension distribution is normal.

88. Let x denote the velocity.

Normal probability plot:



Dotplots:



It does not show linear trend. Thus, it is not plausible that the population distribution is normal.

Section 5.1

9. a. $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1.$

i.e., $\int_{20}^{30}\int_{20}^{30} K(x^2+y^2)\,dx\,dy$

$= K\int_{20}^{30}\left(\frac{19000}{3}+10y^2\right)dy$

$= K\cdot\frac{380000}{3} = 1.$

Thus $K = \frac{3}{380000}$

b. $P(X<26 \text{ and } Y<26) = \int_{20}^{26}\int_{20}^{26} \frac{3}{380000}(x^2+y^2)\,dx\,dy$

$= \frac{3}{380000}\int_{20}^{26}(3192+6y^2)\,dy$

$= \frac{3}{380000}\times 38304$

$= 0.3024$

c. We want to find $|X-Y| \le 2$. That is.

$|x-y| \le 2$. That is, $x-2 \le y \le x+2$.

for $x \in [20,30]$, $y \in [20,30]$.

Thus the total integral region is

$$\int_{20}^{22}\int_{20}^{x+2} f(x,y)\,dy\,dx + \int_{22}^{28}\int_{x-2}^{x+2} f(x,y)\,dy\,dx$$

$$+\int_{28}^{30}\int_{x-2}^{30} f(x,y)\,dy\,dx$$

$$= \frac{3}{380000}\left[\int_{20}^{22}\int_{20}^{x+2}(x^2+y^2)\,dy\,dx + \int_{22}^{28}\int_{x-2}^{x+2}(x^2+y^2)\,dy\,dx\right.$$

$$\left. + \int_{28}^{30}\int_{x-2}^{30} f(x,y)\,dy\,dx\right]$$

$$= \frac{3}{380000} \times \frac{136544}{3}$$

$$= 0.3593$$

d. $f_X(x) = \int_{20}^{30} f(x,y) \cdot dy$

$= \frac{3}{380000} \int_{20}^{30} (x^2+y^2)\, dy$

$= \frac{1}{20} + \frac{3}{38000} x^2$

e. Since $f_Y(y)$ is symmetric about $f_X(x)$.

$f_Y(y) = \frac{1}{20} + \frac{3}{38000} y^2.$

$f(x,y) = \frac{3}{380000}(x^2+y^2) \neq f_X(x) \cdot f_Y(y).$

Thus X and Y are not independent.

12.

a. $P(X \geq 3) = \int_3^{\infty} \int_0^{\infty} x e^{-x(1+y)}\, dy\, dx$

$= \int_3^{\infty} (e^{-x})\, dx$

$= e^{-3}$

$= 0.0498$

b. $f_X(x) = \int_0^\infty f(x,y)\,dy = \int_0^\infty x e^{-x(1+y)}\,dy$

$= \left[-e^{-x(1+y)}\right]_0^\infty$

$= e^{-x}$

$f_Y(y) = \int_0^\infty f(x,y)\,dx = \int_0^\infty x e^{-x(1+y)}\,dx$

$= \left[-\frac{1}{1+y}x - \frac{1}{(1+y)^2}\right] e^{-(1+y)x} \Big|_0^\infty$

$= \frac{1}{1+y^2}$

Since $f(x,y) = x e^{-x(1+y)} \neq f_X(x) \cdot f_Y(y)$.

they are not independent.

c. We first find $P(X \leq 3 \text{ and } Y \leq 3)$.

That is. $\int_0^3 \int_0^3 x e^{-x(1+y)}\,dy\,dx$

$= \int_0^3 (e^{-x} - e^{-4x})\,dx$

$= \frac{3}{4} - \frac{e^{-12}}{4} - e^{-3}$

Thus P(at least one component exceeds 3)

$= 1 - P(X \leq 3 \text{ and } Y \leq 3)$

$= \frac{1}{4} + \frac{e^{-12}}{4} + e^{-3}$

$= 0.2998$

18. a. $P_{Y|X}(0|1) = \frac{p(1,0)}{p_X(1)} = \frac{0.08}{0.08+0.2+0.06} = 0.2353$

$P_{Y|X}(1|1) = \frac{p(1,1)}{p_X(1)} = \frac{0.20}{0.34} = 0.5882$

$P_{Y|X}(2|1) = \frac{p(1,2)}{p_X(1)} = \frac{0.06}{0.34} = 0.1765$

b. $P_{Y|X}(y|2) = \frac{P(2,y)}{P_X(2)} = \frac{p(2,y)}{0.06+0.14+0.30} = \frac{p(2,y)}{0.5}$

when $y=0$

$P_{Y|X}(0|2) = \frac{p(2,0)}{0.5} = \frac{0.06}{0.5} = 0.12$

when $y=1$,

$P_{Y|X}(1|2) = \frac{p(2,1)}{0.5} = \frac{0.14}{0.5} = 0.28$

when $y=2$,

$P_{Y|X}(2|2) = \frac{p(2,2)}{0.5} = \frac{0.30}{0.5} = 0.60$

c. $P(Y\le 1 \mid X=2) = P(Y=0 \mid X=2) + P(Y=1 \mid X=2)$

$= P_{Y|X}(0|2) + P_{Y|X}(1|2)$

$= 0.12 + 0.28$

$= 0.40$

d. $P_{X|Y}(x|2) = \dfrac{p(x,2)}{P_Y(2)} = \dfrac{p(x,2)}{0.02+0.06+0.30} = \dfrac{p(x,2)}{0.38}$

when $x=0$,

$P_{X|Y}(0|2) = \dfrac{p(0,2)}{0.38} = \dfrac{0.02}{0.38} = 0.0526$

when $x=1$,

$P_{X|Y}(1|2) = \dfrac{p(1,2)}{0.38} = \dfrac{0.06}{0.38} = 0.1579$

when $x=2$,

$P_{X|Y}(2|2) = \dfrac{p(2,2)}{0.38} = \dfrac{0.30}{0.38} = 0.7895$

19. a. $f_{Y|X}(y|x) = \dfrac{f(x,y)}{f_X(x)} = \dfrac{\frac{3}{380000}(x^2+y^2)}{\frac{1}{20} + \frac{3}{3800}x^2} = \dfrac{3x^2+3y^2}{19000+300x^2}$

$f_{X|Y}(x|y) = \dfrac{f(x,y)}{f_Y(y)} = \dfrac{\frac{3}{380000}(x^2+y^2)}{\frac{1}{20} + \frac{3}{3800}y^2} = \dfrac{3x^2+3y^2}{19000+300y^2}$

b. $P(Y\ge 25 \mid X=22) = \int_{25}^{30} f_{Y|X}(y\ge 25 \mid 22)\, dy$

$$= \int_{25}^{30} \frac{1452 + 3y^2}{33520} dy$$

$$= \left[\frac{1452}{33520} y + \frac{1}{33520} y^3 \right]_{25}^{30}$$

$$= 0.5551$$

$$P(Y \geq 25) = \int_{25}^{30} \cdot f_Y(y) \cdot dy = \int_{25}^{30} \left(\frac{3}{38000} y^2 + \frac{1}{20} \right) \cdot dy$$

$$= \frac{1}{38000} \left[y^3 + 1900y \right]_{25}^{30}$$

$$= 0.5493$$

Thus $P(Y \geq 25 | X = 22) > P(Y \geq 25)$

c. $$E(Y|X=22) = \int_{20}^{30} \cdot y \cdot f_{Y|X}(y|22) \cdot dy$$

$$= \int_{20}^{30} \frac{1452y + 3y^3}{33520} dy$$

$$= \frac{3}{33520} \int_{20}^{30} (484y + y^3) dy$$

$$= \frac{3}{33520} \left[242y^2 + \frac{1}{4} y^4 \right]_{20}^{30}$$

$$= 25.37$$

$$V(Y|X=22) = E(Y^2|X=22) - [E(Y|X=22)]^2$$

$$E(Y^2 \mid X=22) = \int_{20}^{30} y^2 f_{Y|X}(y \mid 22)\, dy$$

$$= \frac{3}{33520} \int_{20}^{30} (484y^2 + y^4)\, dy$$

$$= \frac{3}{33520} \left[\frac{484}{3} y^3 + \frac{1}{5} y^5 \right]_{20}^{30}$$

$$= 652.03$$

Thus $V(Y \mid X=22) = 652.03 - 25.57^2 = 8.3831$

$\sigma = \sqrt{V(Y \mid X=22)} = 2.89.$